Key Features

Post-quantum security

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Arc's post-quantum roadmap covers wallet signatures, validator authentication, private state, and supporting infrastructure against quantum-era attacks.

Arc's post-quantum roadmap covers wallet signatures, validator authentication, private smart contract state, and offchain infrastructure.

Post-quantum features are on the roadmap and not yet available on Arc.

Why post-quantum security matters

Most public-key cryptography used today is vulnerable to large-scale quantum computers. If those computers become practical, blockchains face two risks:

- **Signature forgery.** A quantum computer that breaks public-key cryptography can forge signatures that secure wallets, authorize transactions, and authenticate network participants.
- **Harvest-now, decrypt-later attacks.** Encrypted data captured today can be stored and decrypted later when quantum attacks become practical, exposing private transaction details, balances, and other sensitive data.

Because blockchain data is long-lived, post-quantum protections need to be in place before quantum attacks are widely available.

Post-quantum roadmap

Arc's roadmap phases each layer in a production-aligned sequence.

Milestone	Release target	Scope	Why it matter
Post-quantum wallet signatures	Mainnet launch	Arc mainnet launches support for a post-quantum wallet signature scheme using `SLH-DSA-SHA2-128s` .	Classical wallet signatures are the most immediate quantum risk to user funds.
Post-quantum privacy	Near-term	Arc Privacy mainnet introduces post-quantum protections for encrypted state and node-to-node communication in privacy mode.	Private balances, counterparties, and transaction details face harvest-now, decrypt-later risk without post-quantum encryption.
Offchain infrastructure upgrades	Mid-term	Circle upgrades infrastructure such as TLS, encrypted data flows, and related operational systems.	Offchain infrastructure uses the same vulnerable cryptography as onchain systems.
Post-quantum validator signatures	Long-term	Arc adds a quantum-resistant signature scheme for validators.	Consensus signatures protect the integrity of the Arc ledger.

Post-quantum wallet signatures

Arc introduces beta support for a post-quantum wallet signature based on `SLH-DSA-SHA2-128s` at mainnet launch. Adoption is opt-in.

Ecosystem constraints to keep in mind:

- Hardware wallet support will take time to mature.
- Post-quantum standards are still evolving, so long-term signature choices may change.

⚠ Expect a transition period as tooling, wallet support, and integrations mature.

Post-quantum privacy

Arc Privacy addresses the harvest-now, decrypt-later threat. When Arc Privacy launches, encrypted state and private transaction flows use post-quantum cryptography.

Attackers can capture encrypted data today and attempt to decrypt it later when quantum attacks become practical. Post-quantum encryption at the platform level protects sensitive balances, transaction details, and recipients without requiring you to implement custom post-quantum cryptography.

Post-quantum validator signatures

Validator authentication also requires post-quantum protection to keep the ledger resilient. Arc adds post-quantum validator signatures in a later phase.

This sequencing is intentional: validator upgrades must be introduced carefully to preserve throughput, latency, and operational reliability. Because Arc uses sub-second finality, the window to exploit validator signatures is narrower than the wallet-signature risk, making wallets the higher-priority target.

Offchain infrastructure

Quantum resilience extends beyond the chain itself. Circle's roadmap includes upgrading TLS, encrypted data flows, access controls, cloud environments, and other operational systems that support Arc.

Offchain traffic and stored data face the same post-quantum threat vectors as onchain data when vulnerable cryptography remains in use.

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